7.5 POWER PLANT AND RELATED SYSTEMS

7.5.1 **Engines** (see Fig. 7-6)

Two Allison 250-C20 series gas turbine engines are located in separate fireproof compartments aft of the main transmission and above the cargo compartment. Each engine is divided into four modules:

- Compressor section consisting of a compressor support case assembly, a compressor and a diffuser assembly. The compressor has six axial stages and one centrifugal stage. Components mounted on the compressor section are an anti-icing air valve and a compressor bleed control valve.

- Combustion section consisting of an outer combustion case and a combustion liner. A spark igniter and a fuel nozzle are mounted in the aft end of the outer combustion case. Two external air ducts connect the compressor section with the combustion section.

- Turbine section consisting of turbine supports, collector supports, a gas producer turbine rotor and a power turbine rotor. The two stage gas producer turbine drives the compressor and N_1 accessory gear train. The two-stage power turbine furnishes the output power of the engine via the power and accessory gearbox. Mounted on the turbine section are a thermocouple harness, exhaust ducts and a fire-shield to which are attached the two fire detectors.

- Power and accessory gearbox section housing the main power and accessory gear drive trains in a single gear case. The gear case serves as the structural support of the engine. All engine driven components including the engine-mounted accessories are attached to the case. A two-stage gear set reduces the power turbine rotational speed to the output drive rotational speed. Accessories driven by the power turbine (N_2) are the N_2 tachgenerator and N_2 governor. The gas producer gear train drives the oil pump assembly (internally mounted), the fuel pump, N_1 tachgenerator, gas producer fuel control and starter/generator.

7.5.2 **Air Induction**

The engines receive air from the main transmission compartment which acts as a plenum chamber. An air intake fairing above the cabin roof forward of the transmission supplies a constant airflow towards the transmission compartment.

EFFECTIVITY *0451-9999, and 0001-0450 After SB 60-37*

A deflector shield is installed within the air intake fairing to direct the airflow downwards thus preventing injestion of foreign matter into the engines.

EFFECTIVITY *ALL*

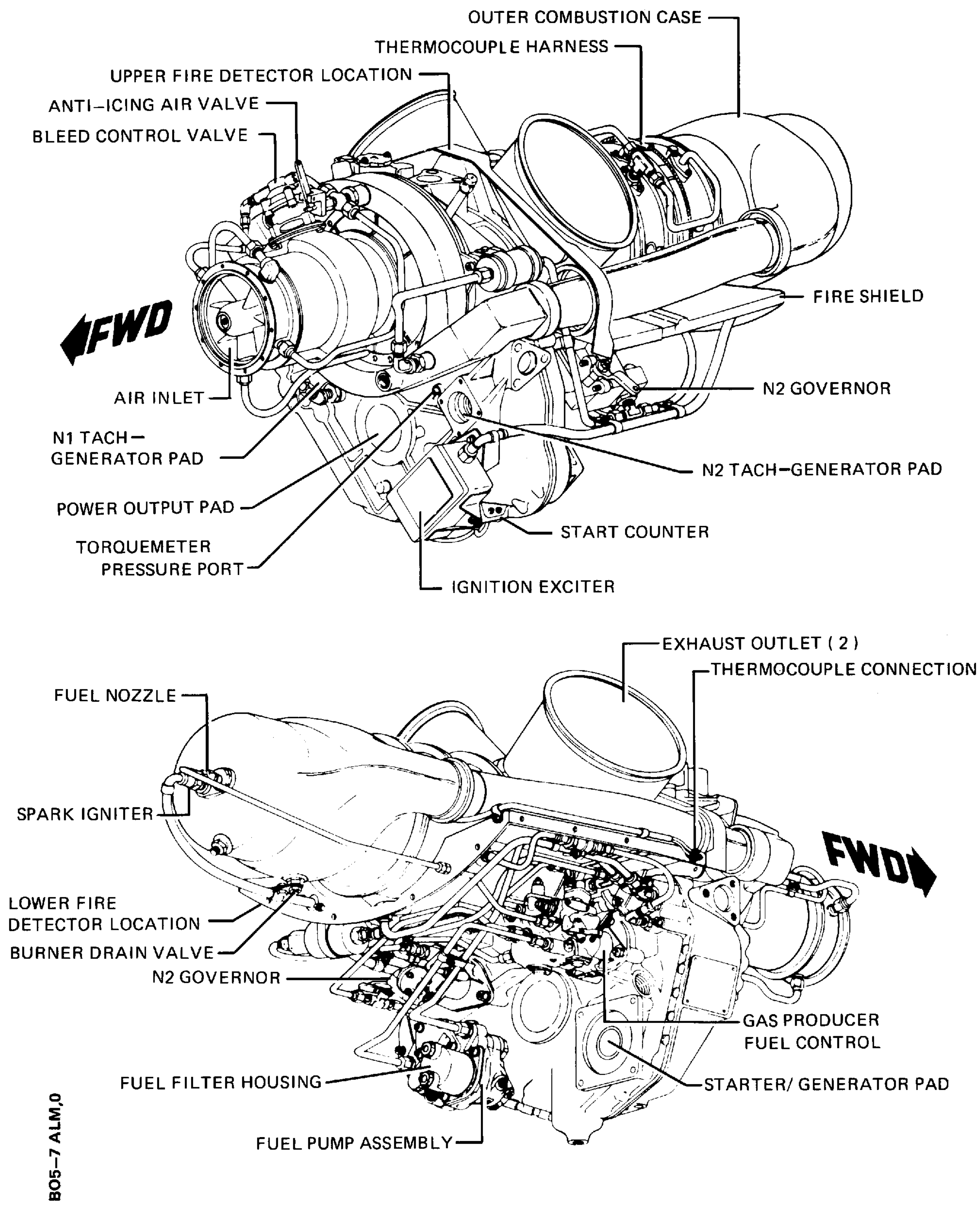


Fig. 7-6 Engine Component Locations

7.5.3 Inlet Anti-icing System

To accomplish engine anti-icing functions, hot compressor discharge air is tapped from the diffuser scroll through an electrical actuator operated anti-icing valve and routed via external lines to the compressor inlet guide vanes and front bearing support hub. The electrical actuators are controlled by the ENG ANTI-ICING - 1, 2 switches and monitored by the **ANTI ICING 1 (2)** caution lights all located on the center console switch panel.

In the ON position, the ENG ANTI-ICING switches supply power from the Main Bus via ENG ANTI-ICING - 1, 2 circuit breakers to the respective actuator motor retracting the actuator arm and opening the anti-icing valve.

A limit switch stops the actuator motor and activates the respective **ANTI ICING** caution light. The caution light remains on until after the system is switched OFF and the actuator has fully closed the anti-icing valve.

7.5.4 Fuel and Control System

Each engine fuel and control system comprises an engine-mounted fuel pump and filter assembly, a gas producer fuel control, a power turbine (N_2) governor, and a fuel nozzle. This system controls engine power output by controlling the gas producer speed. Gas producer speed levels are established by the action of the power turbine fuel governor which senses power turbine (N_2) speed. The power turbine (load) speed is selected by the pilot (ENGINE TRIM switch) and the power required to maintain this speed is automatically maintained by power turbine governor action on metered fuel flow.

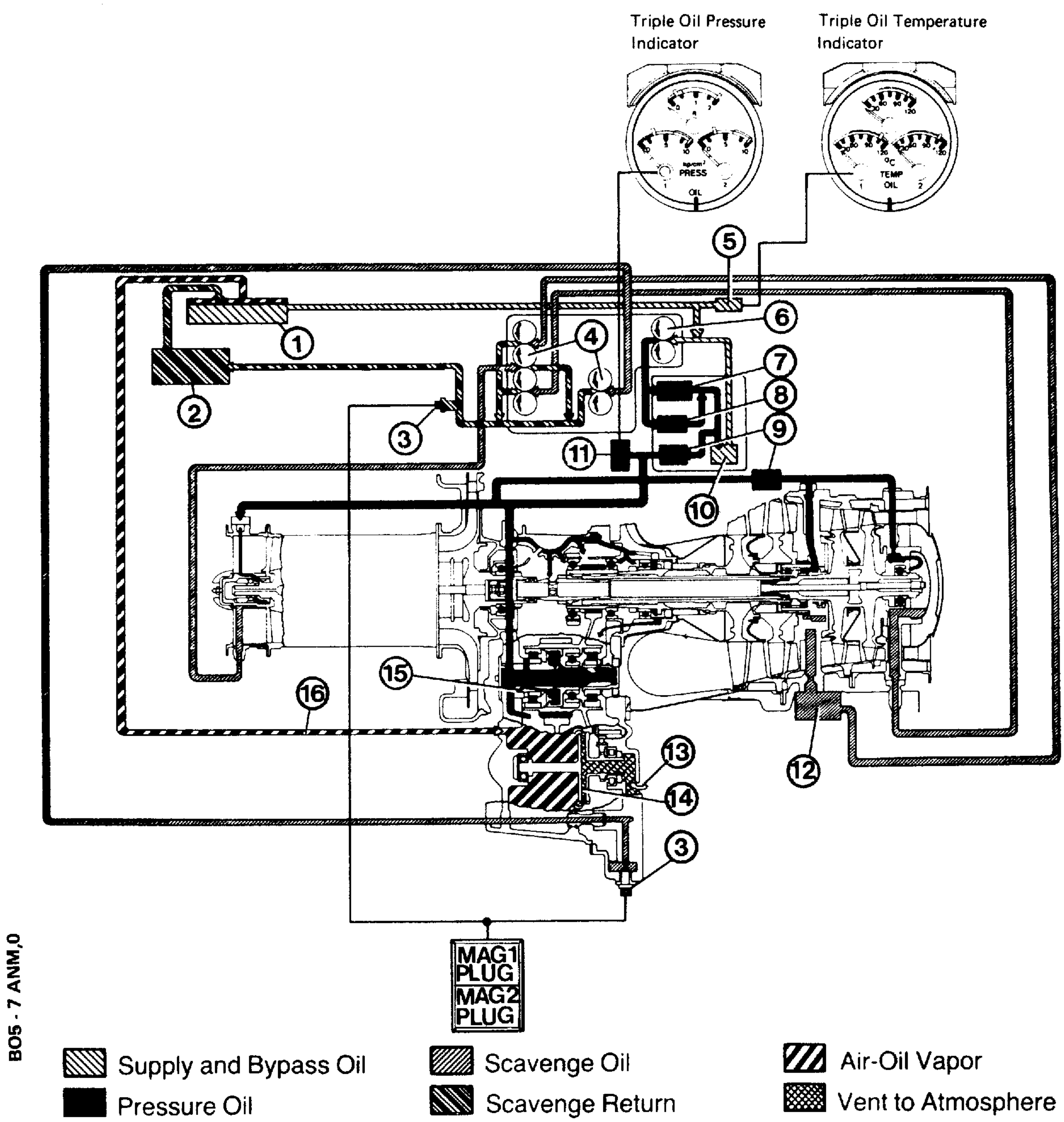
A differential pressure switch is connected to the engine fuel filter to detect the beginning of clogging and activate the respective **FILT 1 (2)** warning light. A clogged filter element will be bypassed automatically supplying unfiltered fuel to that engine.

7.5.5 Oil System (see Fig. 7-7)

Each engine is lubricated by an independent pressurized recirculating dry sump system. A gear-type pressure and scavenge pump assembly is mounted within the gearbox. An assembly containing an oil filter element, a filter bypass valve, and a pressure regulating valve is located in the upper RH side of the gearbox housing and is accessible from the top of the engine.

A combined dual engine oil tank and engine/transmission oil cooler assembly is located to the right of the main transmission. The forward half of the tank/cooler assembly is connected to the left engine oil system while the rear half is connected to the right engine oil system. Two engine-mounted indicating chip detectors that activate the respective **MAG PLUG 1 (2)** warning light, are installed in each system.

The operation of each oil system is completely automatic self-regulating and does not require operator control beyond monitoring the indicators.



1. OIL TANK
2. OIL COOLER
3. MAGNETIC CHIP DETECTORS (2)
4. SCAVENGE PUMPS
5. OIL TEMPERATURE SENSOR
6. HIGH PRESSURE PUMP
7. OIL FILTER
8. OIL FILTER BYPASS VALVE
9. CHECK VALVES
10. PRESSURE REGULATING VALVE
11. OIL PRESSURE TRANSMITTER
12. EXTERNAL SUMP
13. OVERBOARD BREATHER (TO EXHAUST)
14. AIR/OIL SEPARATOR
15. TORQUEMETER PRESSURE (TO INDICATOR)
16. OIL TANK VENT LINE

Fig. 7-7 Engine Oil System Schematic

7.5.6 **Starting and Ignition System**

Each engine is equipped with an independent starting/ignition system for starting the engines individually and so designed that simultaneous starting of both engines is impossible. A starter/generator unit mounted to the accessory gearbox drives the N_1 gear train when the starter motor circuits are activated. An ignition exciter unit and spark igniter comprise the ignition portion of each engine's system.

The STARTER/GENERATOR - ENG 1, ENG 2 switches on the center console switch panel are spring-loaded to the center off position when using the START position. In the START position, the switch supplies electrical power from the Main Bus via a common STARTERS circuit breaker to the respective starter relay. This relay in turn activates the starter motor circuit by supplying power from the Starter Bus and provides power to the respective IGNITION switch.

With the IGNITION switch in the on position, the exciter unit begins sending high power impulses to the spark igniter. Venting the engines is therefore possible by energizing the starter circuit without switching on the ignition circuit.

7.5.7 **Control System**

The engines have a conventional control system, i.e. power lever position and the degree of collective pitch basically establish the power output demands placed on the engines.

The gas producer (N_1) rpm of both engines is independently controlled by means of separate power levers mounted on the cockpit ceiling forward of the overhead console. Each power lever is mechanically connected to its respective engine gas producer fuel control to permit manual establishment of the gas producer speeds between OFF, IDLE and FLIGHT.

A separate positive lock is provided for each lever in the IDLE position so that repositioning of the levers to a position other than IDLE requires depressing the lock buttons.

Individual and simultaneous dual engine power (N_2) trimming is accomplished by means of a four-way ENGINE TRIM switch on each collective lever switch box (see Fig. 7-14). Pressing the switch to either left or right of center will increase the respective engine torque and simultaneously reduce the torque of the other engine. Pressing the switch either forward or aft will simultaneously increase or decrease respectively the torque of both engines. The pilot's ENGINE TRIM switch has command authority over the copilot's switch if installed.

A mechanical connection between the N_2 governor trim actuator and the collective control hydraulic boost actuator provides automatic rpm control (droop compensation) when changes in collective pitch occur.

7.5.8 **Indicators** (see Fig. 7-3)

Torque indicator (% TORQUE) simultaneously indicates the percent of torque output of both engines. Each engine's torquemeter, incorporated in the accessory gearbox, provides an oil pressure signal directly proportional to the output torque. This oil pressure is piped directly to the respective indicator.

Triple tachometer indicator (ENG-ROTOR PERCENT RPM) simultaneously indicates both engine's power turbine speed (N_2) on an outer scale in percent rpm. The engine % N_2 rpm signals are produced by a tachgenerator mounted on the left front face of the accessory gearbox.

Turbine outlet temperature indicators (˚C x 100 TOT) indicate individual engine exhaust gas temperature measured by four thermocouple probes placed downstream of the power turbine. The bimetallic thermocouples generate DC voltage directly proportional to the sensed gas temperature.

Gas producer tachometers (% N_1) indicate individual engine gas producer speed (N_1) in percent of rated rpm. The gas producer turbine speed is measured by a tachgenerator mounted on the right front face of the accessory gearbox.

Triple oil temperature indicator (˚C TEMP OIL) simultaneously indicates engine 1 oil temp (left scale) and engine 2 oil temp (right scale). Each engine's oil temperature sensor is installed in an oil supply line fitting mounted on the top right front face of the accessory gearbox. The triple oil temperature indicator is powered from the Main Bus via a fuse.

Triple oil pressure indicator (kp/cm^2 PRESS OIL) simultaneously indicates engine 1 oil pressure (left scale) and engine 2 oil pressure (right scale). Each engine oil pressure transmitter is fitted to a clamp securing the oil lines to the engine compartment floor. The triple oil pressure indicator is powered from the Main Bus via a fuse.

7.5.9 **Annunciators**

MAG PLUG 1, MAG PLUG 2 warning lights will activate when excessive metallic particles are detected by either engine-mounted chip detector.

ANTI ICING 1, ANTI ICING 2 caution lights will activate when each engine anti-icing system valve is open.